Decoding Drug Resistance in TB

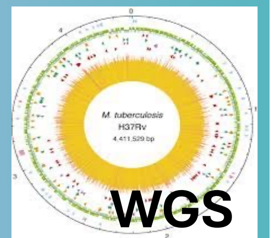
From Cartridge to Genome

INSTITUTE OF TROPICAL MEDICINE ANTWERP

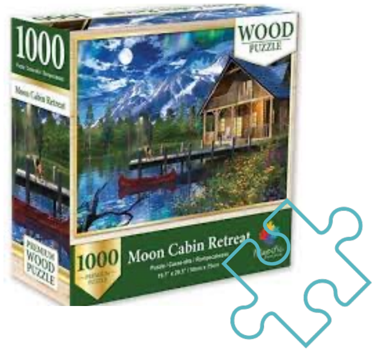
University of Antwerp
Faculty of Medicine and Health Sciences

MOLECULAR ASSAYS FOR TUBERCULOSIS: increased levels of complexity

INFORMATION	*Mtb*/no *Mtb* RIF	*Mtb*/no *Mtb* RIF INH	*Mtb*/no *Mtb* 15 drugs Genotype	*Mtb*/no *Mtb* all drugs Genotype Transmission
TARGETS	MTB-specific 1 drug-target	MTB-specific 3 drug-targets	MTB-specific 18 drug-targets	4.4Mb All drug-targets
SAMPLE TYPE	Raw specimen	Treated specimen	Treated Specimen ***	Cultured isolate
MOLECULAR ASSAY	**Simple NAATs**	**Moderate complexity NAATs**	DEEPLEX **tNGS**	**WGS**

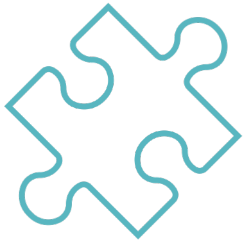


WGS DATA ANALYSIS APPROACHES



Reference mapping approach

- Known genome sequence of closely related organism
- Analysis is less resource-intensive
- Limited to known genomic regions



***De novo* assembly**

- Limited knowledge of genome sequence or closely related organism
- Analysis is resource-intensive
- Describe novel genomic structures
- Species characterisation

ONT WGS ANALYSIS USING GALAXY

Mycobacterium tuberculosis clinical isolates

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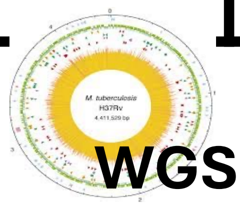
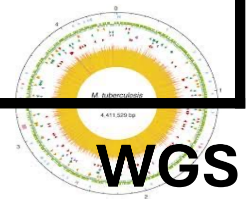
NGS DATA ANALYSIS TOOLS/PLATFORMS: increased levels of complexity

On/off line tools	Galaxy	Command line
• Automated • No installation • User friendly - GUI • Cloud-based • Internet connection needed* • Limited customizability • Runs CLI tools 'under the hood' • Potential data security issues	• Data analysis workflow platform • Hybrid interface • Ease of use of GUI • Flexibility of command line • Local or public server	• Highly customizable • Can be tedious • Needs computing recourses • Needs some training/familiarity with the software • Automation requires programming expertise

DEEPLEX

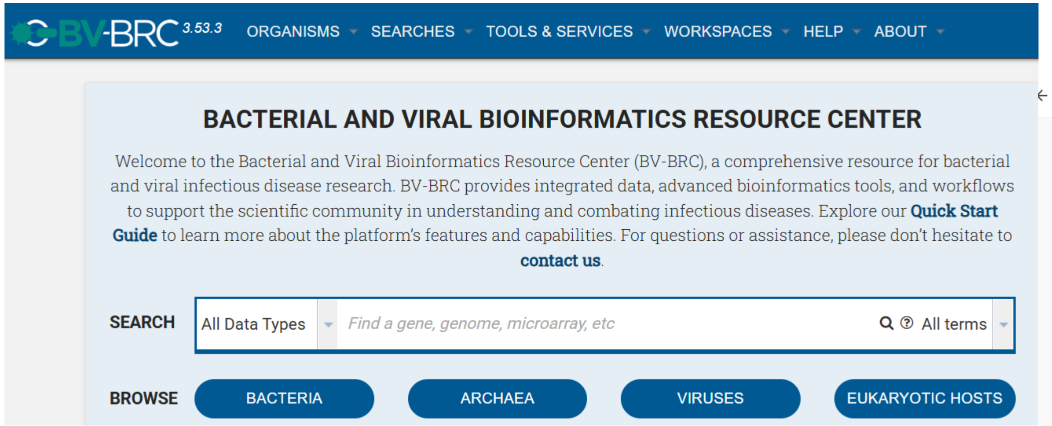




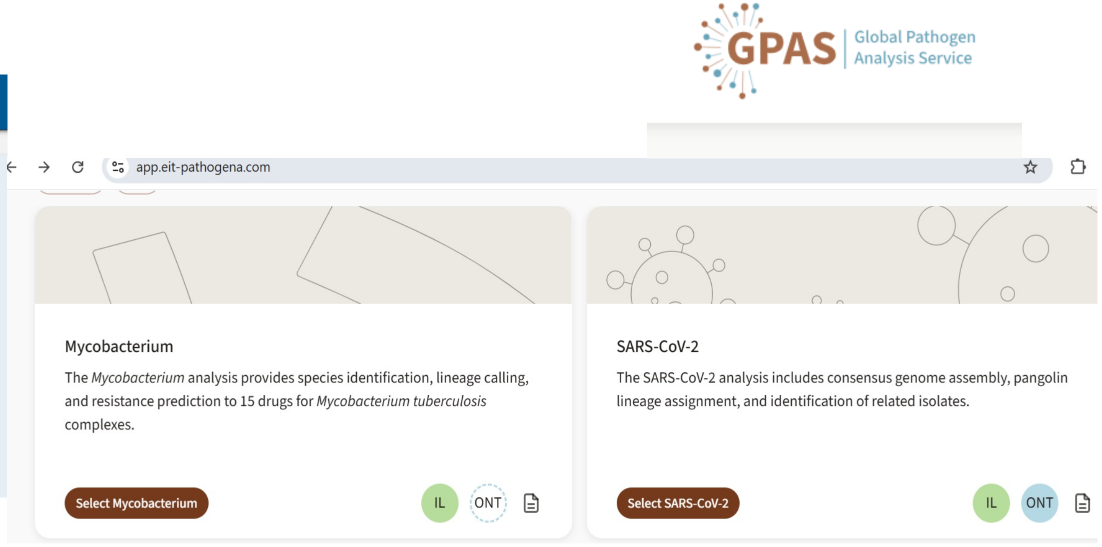
Galaxy PROJECT

From programming languages to user-friendly genomics data analysis platform

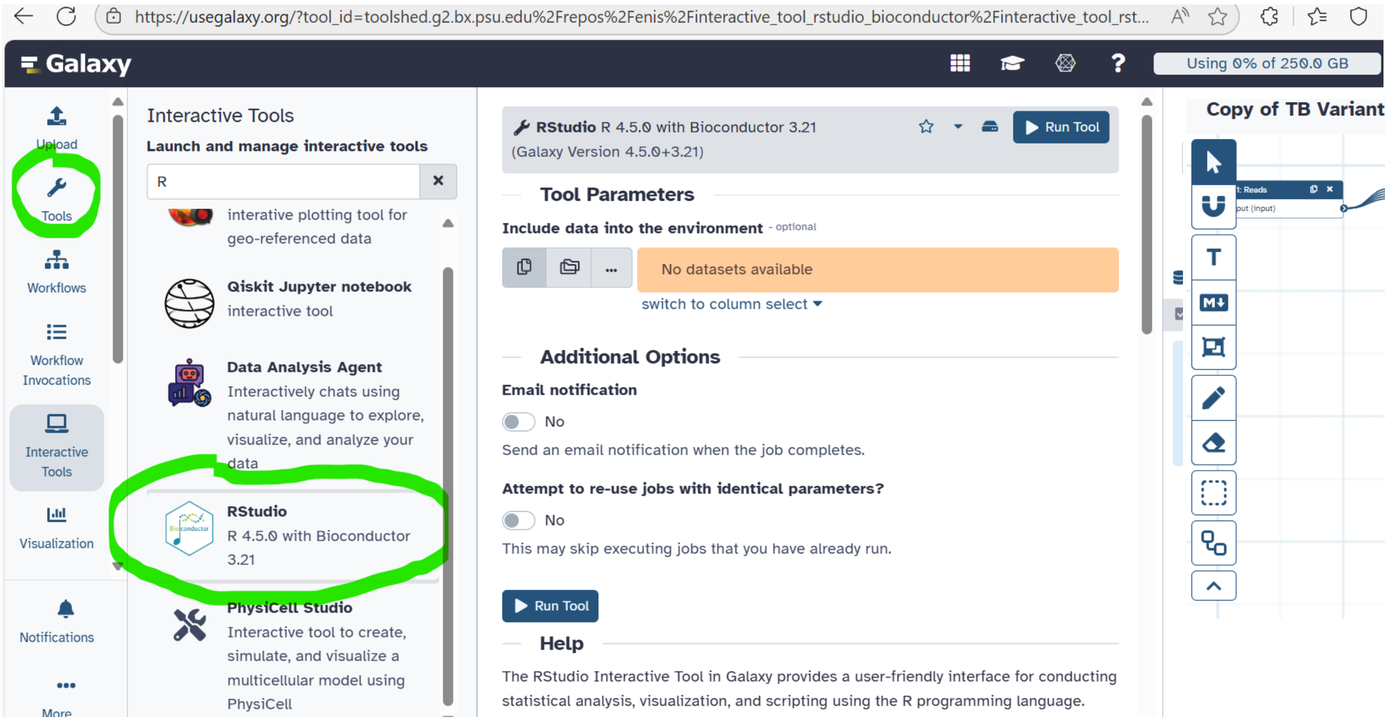
https://www.bv-brc.org/

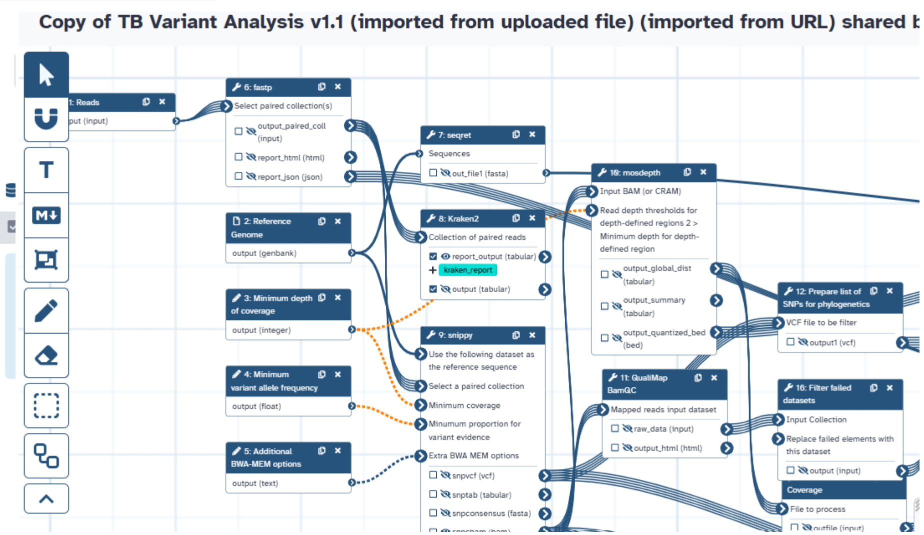


https://app.eit-pathogena.com/



From programming languages to user-friendly genomics data analysis platforms





WGS ANALYSIS FOR *Mtb*

Comprehensive:

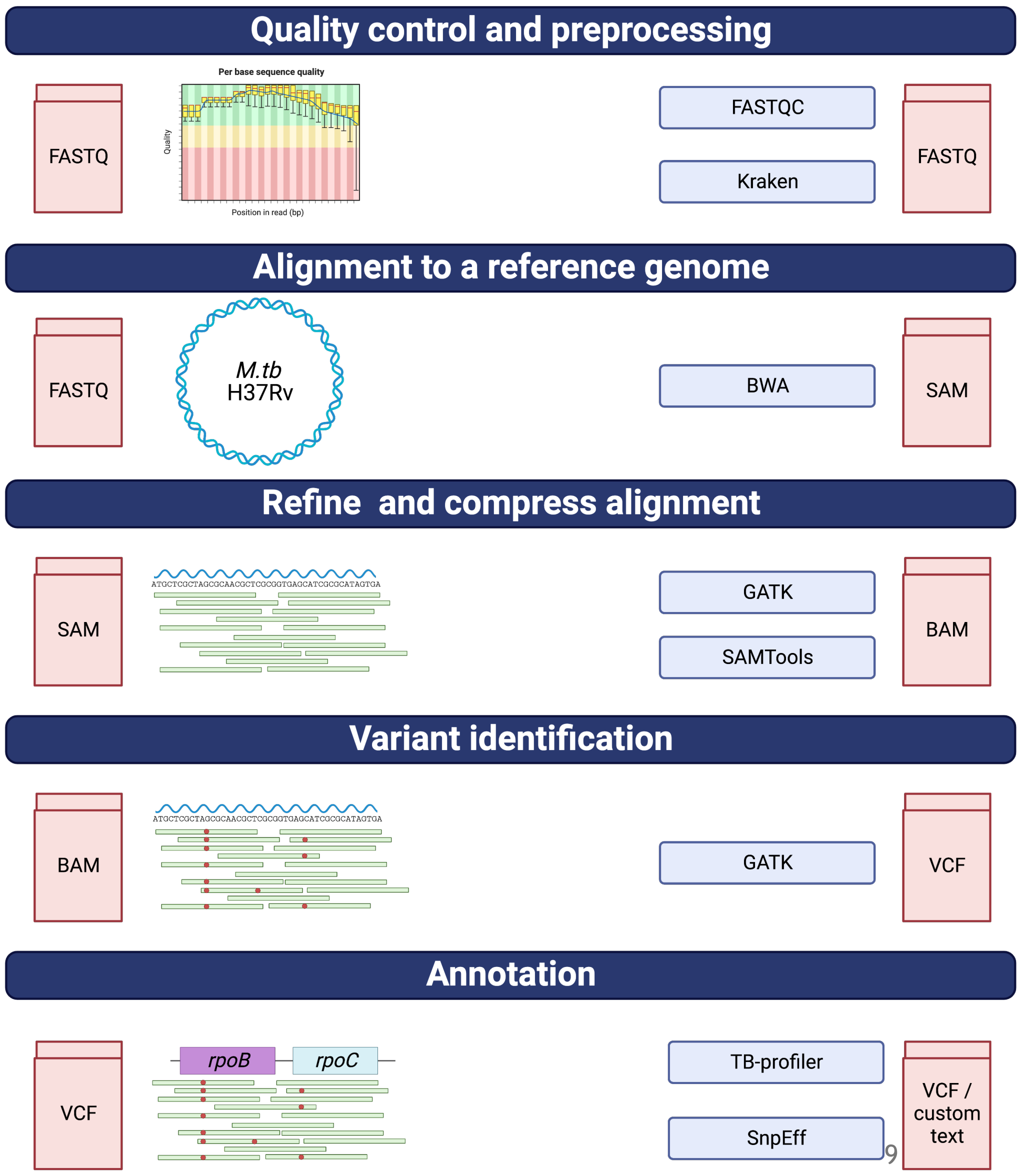
- **QC**
- **Drug resistance**
- **Strain identification**
- **Genome-wide variants**
- **Transmission clusters**
- **Phylogeny**

Focused:

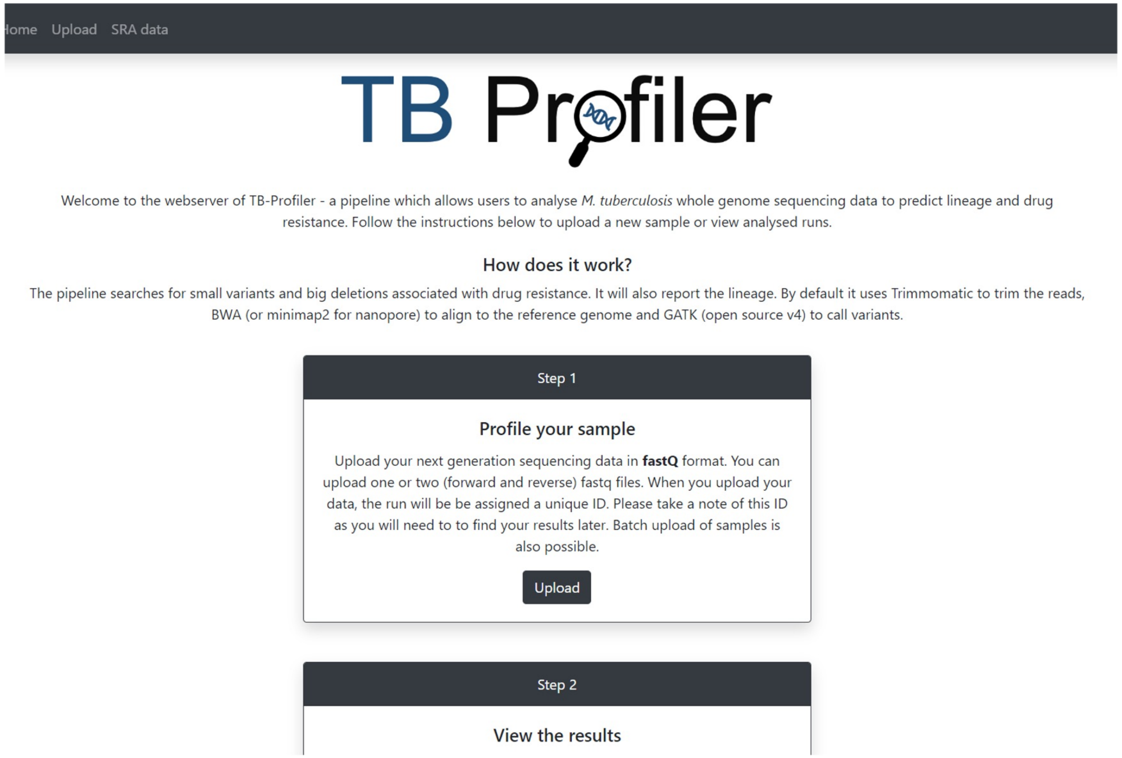
TB Profiler

- **Drug resistance**
- **Strain identification**
- **Basic stats for QC**

•10.1099/mgen.0.000689



TB-Profiler: web-based



- Limited customisation
- One isolate at a time
- Access to 16,000 publicly available sequences
- User-friendly and no command line expertise needed
- Internet connection needed

https://tbdr.lshtm.ac.uk/

TB-Profiler: the command line

```
Usage: tb-profiler profile [-h] [--read1 READ1] [--read2 READ2] [--bam BAM] [--fasta FASTA] [--vcf VCF] [--platform {illumina,nanopore,pacbio}] [--db DB]
                           [--external_db EXTERNAL_DB] [--prefix PREFIX] [--csv] [--txt] [--text_template TEXT_TEMPLATE] [--docx] [--docx_template DOCX_TEMPLATE]
                           [--add_columns ADD_COLUMNS] [--dir DIR] [--depth DEPTH] [--af AF] [--strand STRAND] [--sv_depth SV_DEPTH] [--sv_af SV_AF] [--sv_len SV_LEN]
                           [--mapper {bwa,minimap2,bowtie2,bwa-mem2}] [--caller {bcftools,gatk,freebayes,pilon,lofreq}] [--calling_params CALLING_PARAMS]
                           [--kmer_counter {kmc,dsk}] [--suspect] [--spoligotype] [--update_phylo] [--call_whole_genome] [--snp_dist SNP_DIST] [--no_trim]
                           [--no_flagstat] [--no_clip] [--no_delly] [--threads THREADS] [--ram RAM] [--logging {DEBUG,INFO,WARNING,ERROR,CRITICAL}] [--temp TEMP]
                           [--version]
```

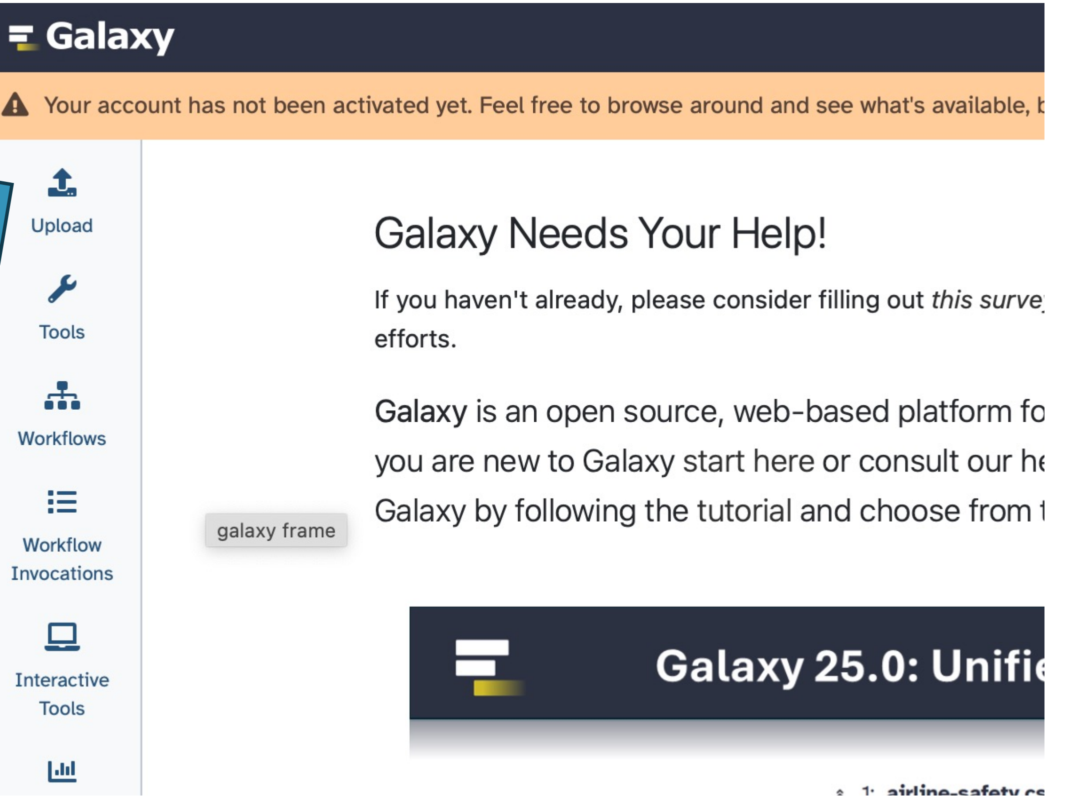
```
Input Options:
  --read1, -1 READ1     First read file (default: None)
  --read2, -2 READ2     Second read file (default: None)
  --bam, -a BAM         BAM file (make sure it has been generated using the H37Rv genome (GCA_000195955.2)) (default: None)
  --fasta, -f FASTA     Fasta file (default: None)
  --vcf, -v VCF         VCF file (default: None)
  --platform, -m {illumina,nanopore,pacbio}
                        NGS Platform used to generate data (default: illumina)
  --db DB               Mutation panel name (default: tbdb)
  --external_db EXTERNAL_DB
                        Path to db files prefix (overrides --db parameter) (default: None)

Output Options:
  --prefix, -p PREFIX   Sample prefix for all results generated (default: tbprofiler)
  --csv                 Add CSV output (default: False)
  --txt                 Add text output (default: False)
  --text_template TEXT_TEMPLATE
                        Jinja2 formatted template for output (default: None)
  --docx                Add docx output (default: False)
  --docx_template DOCX_TEMPLATE
                        Supply custom template for --docx output (default: None)
  --add_columns ADD_COLUMNS
                        Add additional columns found in the mutation database to the text and csv results (default: None)
  --dir, -d DIR         Storage directory (default: .)
```

- **Flexible**
- **Fully customisable**
- **Parallel processing**
- **Internet only needed for initial installation and downloading publicly available genomes**
- **Updated version immediately available**

MTB ONT WGS analysis on Galaxy with TBProfiler

- Base calling has been done: FAST5 → FASTQ

- **UPLOAD DATA TO GALAXY:**

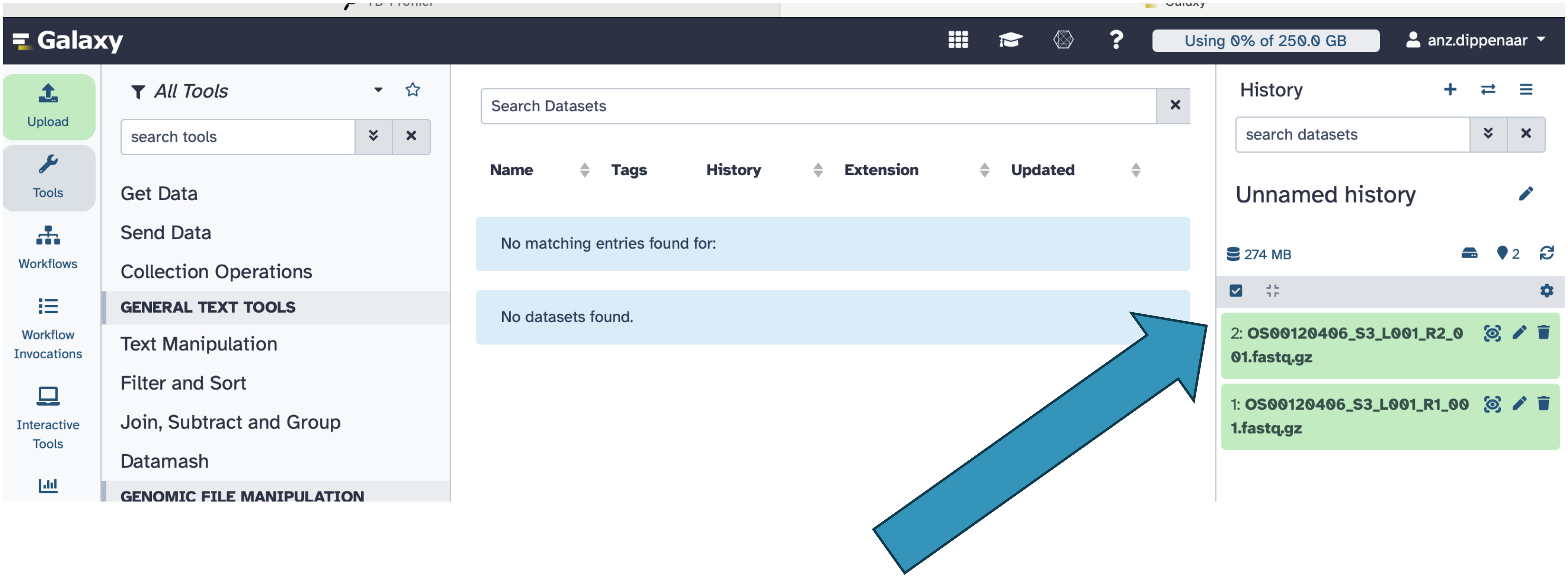


ONT FASTQ file

/usr/local/share/MACMA/tbprofiler/ONT/

ERR9030283.fastq.gz

UPLOADED DATA



PRACTICAL USING GALAXY

- STEPS FOR ANALYSIS:
  - Step 1: Search for TB-profiler in the Galaxy toolbox
  - Step 2: Customise the tool parameters
  - Step 3: Run TB-profiler on **one strain**
  - ...wait for the results
  - Step 4: Interpret the results

CUSTOMISING TB-Profiler PARAMETERS

- Choose the tool parameters appropriate to the analysis needed:
  - **Platform**: Nanopore
  - **Input File Type**: Single Fastq
  - **Output format**: text (the report in text format is more detailed compared to the slimmed-down, easy-to-interpret PDF report)
- The minimum depth and reporting allele frequencies can be customised under "Select advanced options" – 'Yes'

TB-Profiler - GALAXY

- Set parameters
- Select dataset
- “Run Tool”
- Wait…

UNDERSTANDING TB-Profiler OUTPUTS

- Strain type
- Drug resistance profile
- Median depth
- Resistance to individual drugs

```
TBProfiler report
=================

The following report has been generated by TBProfiler.

Summary
-------
ID: strainA
Date: Thu Sep  7 06:31:41 2023
Strain: lineage4.3.2.1
Drug-resistance: MDR-TB
Median Depth: 71

Lineage report
--------------
Lineage Estimated Fraction      Spoligotype     Rd
lineage4        1.000   LAM;T;S;X;H     None
lineage4.3      1.000   mainly-LAM      None
lineage4.3.2    1.000   LAM3    None
lineage4.3.2.1  1.000   LAM3    RD761



Resistance report
-----------------
Drug    Genotypic Resistance    Mutations
Rifampicin      R       rpoB p.Thr400Ala (0.94), rpoB p.Ser450Leu (0.96)
Isoniazid       R       katG p.Ser315Thr (0.93)
Ethambutol      R       embB p.Met306Ile (0.82)
Pyrazinamide    R       pncA p.Lys96Thr (0.91)
Streptomycin    R       rpsL p.Lys88Gln (0.90)
Fluoroquinolones
Moxifloxacin
Ofloxacin
Levofloxacin
Ciprofloxacin
Aminoglycosides
Amikacin
Capreomycin
Kanamycin
Cycloserine
Ethionamide
Clofazimine
Para-aminosalicylic_acid
Delamanid
Bedaquiline
Linezolid
```

TB Profiler for multiple samples (commandline)

- Look at Excel summary on VM:
  /usr/local/share/MACMA/tbprofiler/

`EXAMPLE_TB-profiler_summary_20230907`

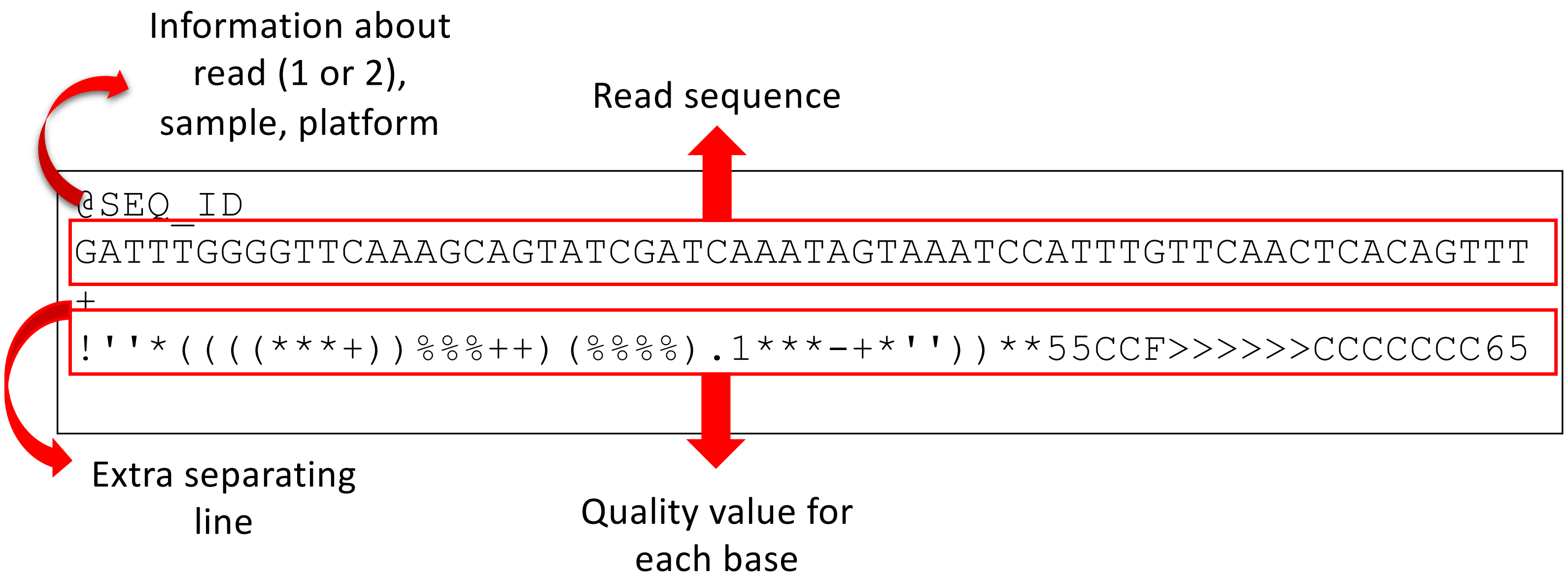
What kind of raw data do we get?

om Illumina sequencers:

- FastQ format: https://en.wikipedia.org/wiki/FASTQ_format

```
@SEQ_ID
GATTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTTGTTCAACTCACAGTTT
+
!''*((((***+))%%%++)(%%%%).1***-+*''))**55CCF>>>>>>CCCCCCC65
```

FastQ format



Phred Quality values

- The quality score of a base, also known as a Phred or Q score, is an integer value representing the estimated probability of an error, i.e. that the base is incorrect. If P is the error probability, then:
- $P = 10^{-Q/10}$
- $Q = -10 \log_{10}(P)$
- Translating to:

Q of 30 = error probability of 1 in 1000
Q of 20 = error probability of 1 in 100